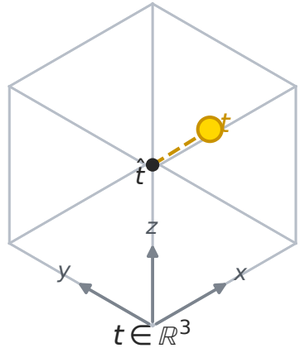
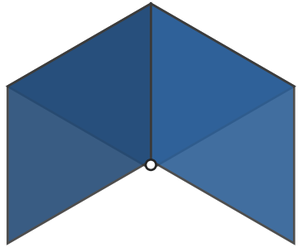
The octree discretization of translation $\mathbb{R}^3$: one equivolumetric partition by construction



camera-frame translation,
anchor $\hat{t}$, extent A

anchor + scale

$t = \hat{t} + Ag$



root cell

box $[\hat{t} - A,\ \hat{t} + A]$,
volume $V_0 = 8 \det A$

subdivide

split each axis

octree split $\times 8$

8 octants $(\pm, \pm, \pm)$,
each $\frac{1}{8}$ the volume

select

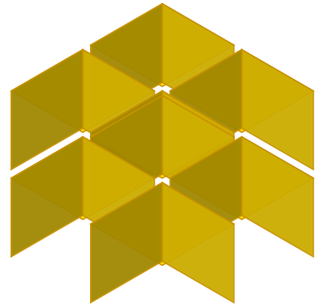
octant index o

chosen octant

equivolumetric:
$t = \hat{t} + A\,g,\quad g \in [-1, 1]^3$

recurse

$A \to A/2$



refine $\times 8$

halve each side,
$N(\ell) = 8^\ell$ cells